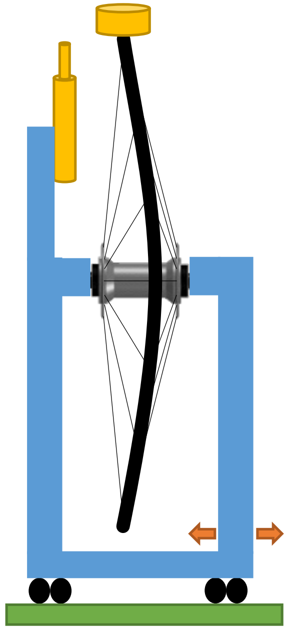
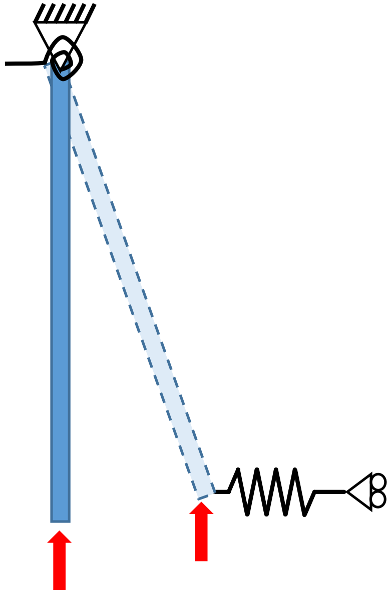
Radial collapse of the bicycle wheel:

Experiments and theory

See embedded comments for additional notes





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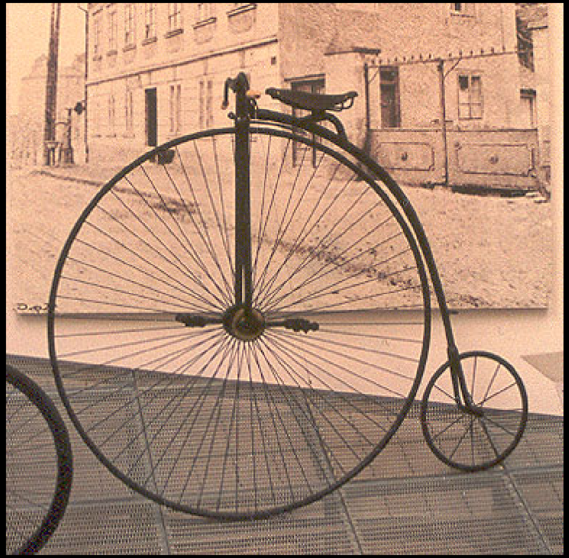
[1] Northwestern McCormick School of Engineering

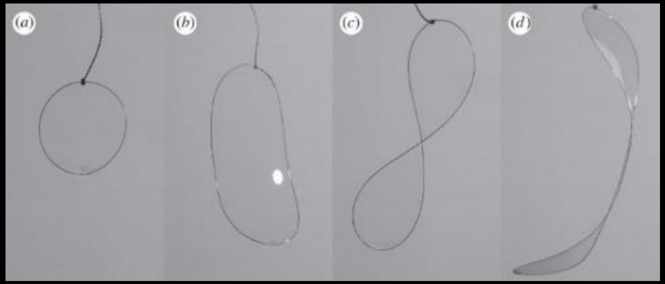
[2] Northeastern University College of Engineering

Tensegrity and instability across size scales

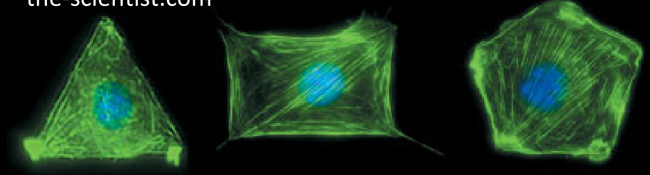
Wikipedia



Giomi, Mahadevan, PRSA *468*(2143)



the-scientist.com



Wikipedia

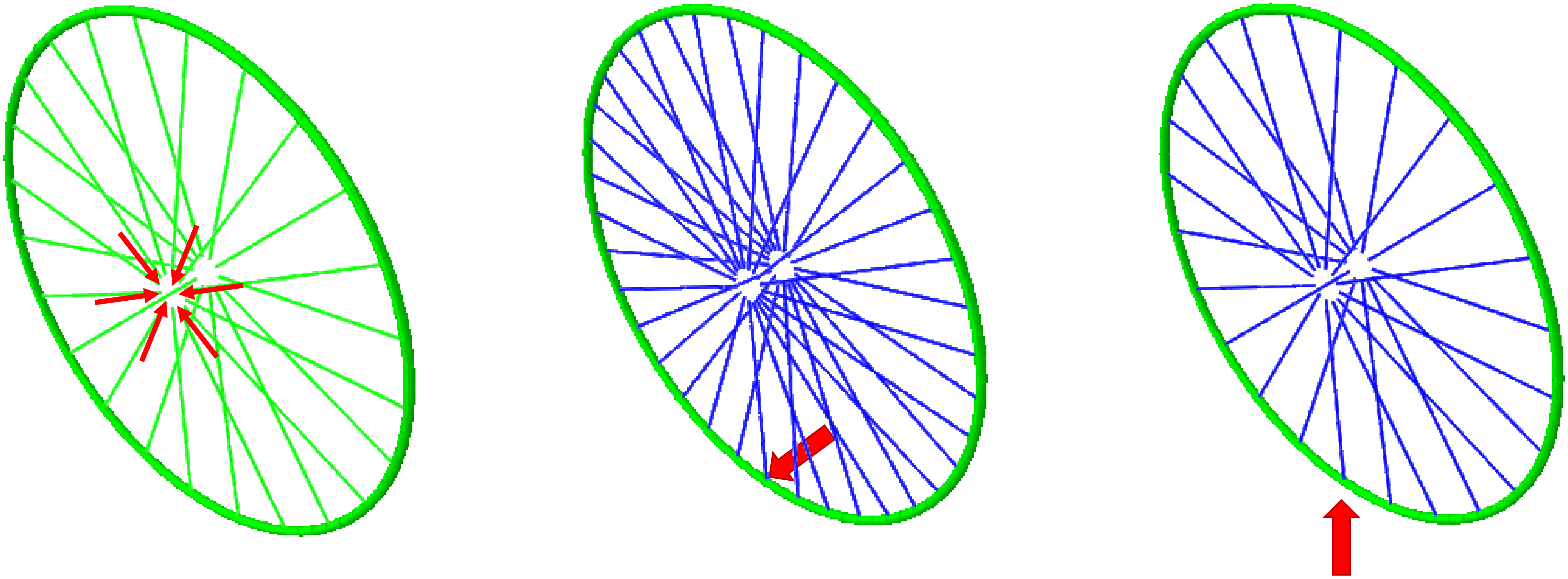
Now *that's* the ultimate wheel pretzel!

www.milanofixed.com/wheel-pretzel/

Wheel failure

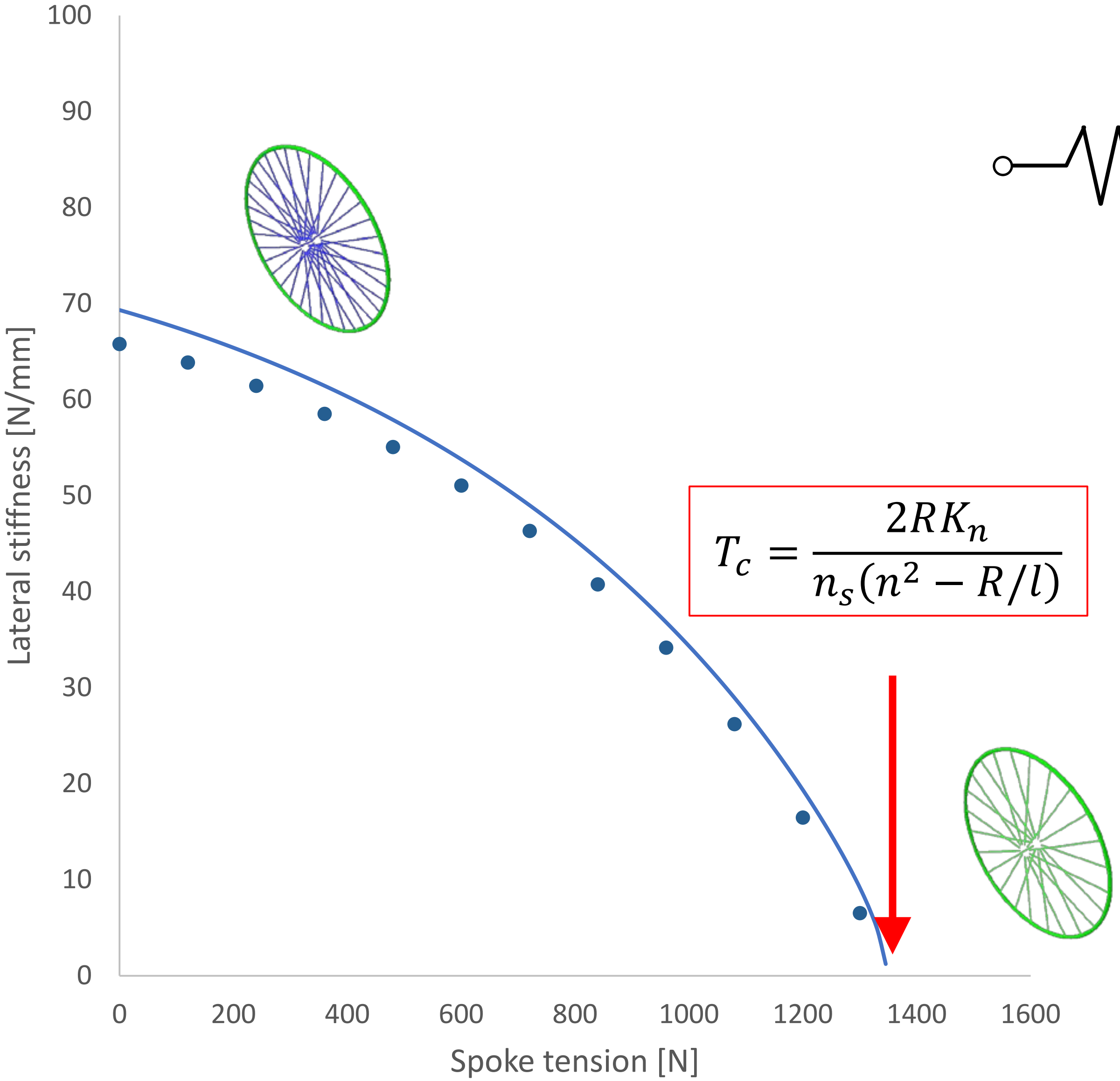
telegraph.co.uk

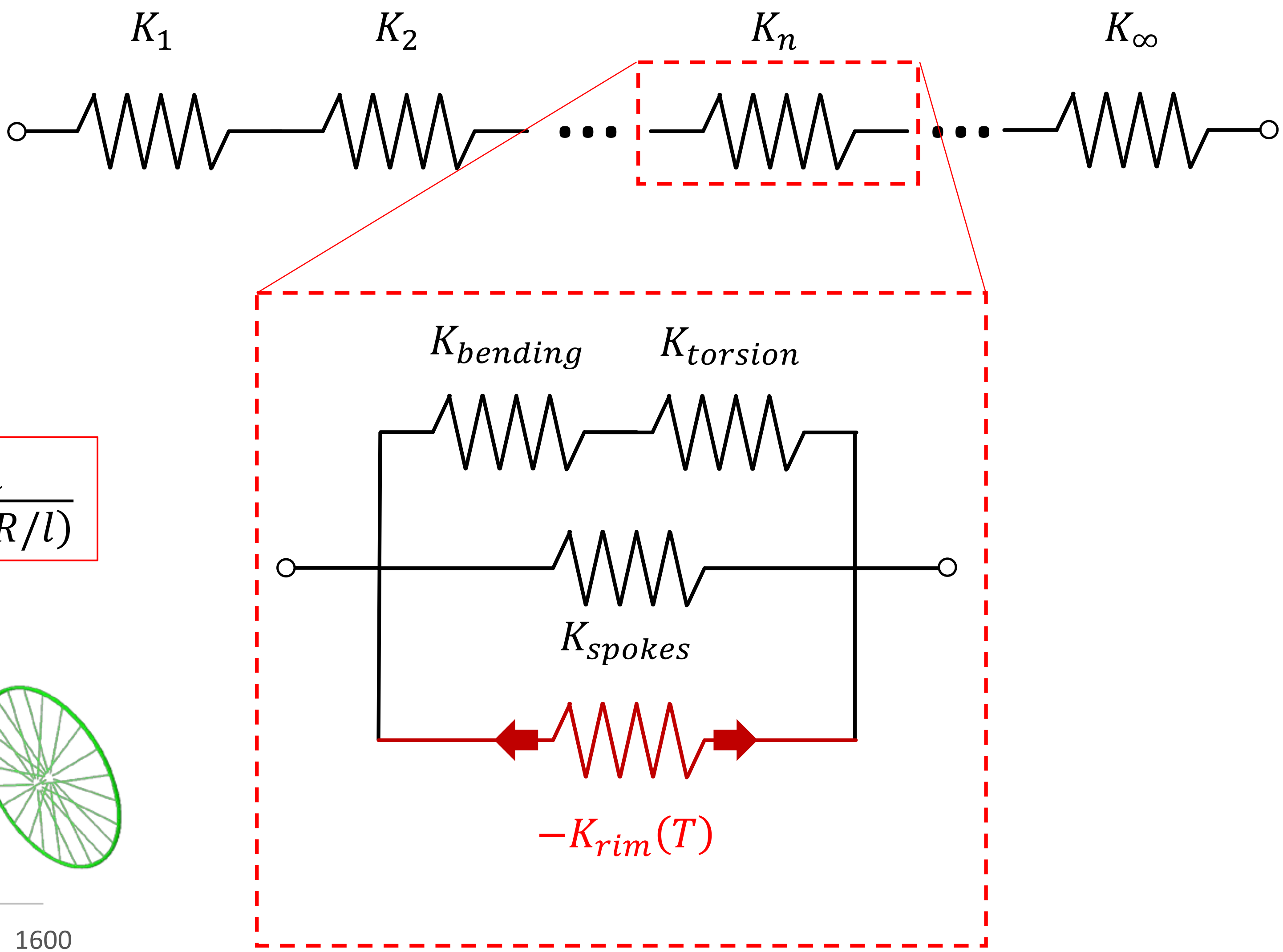
Instabilities of the bicycle wheel



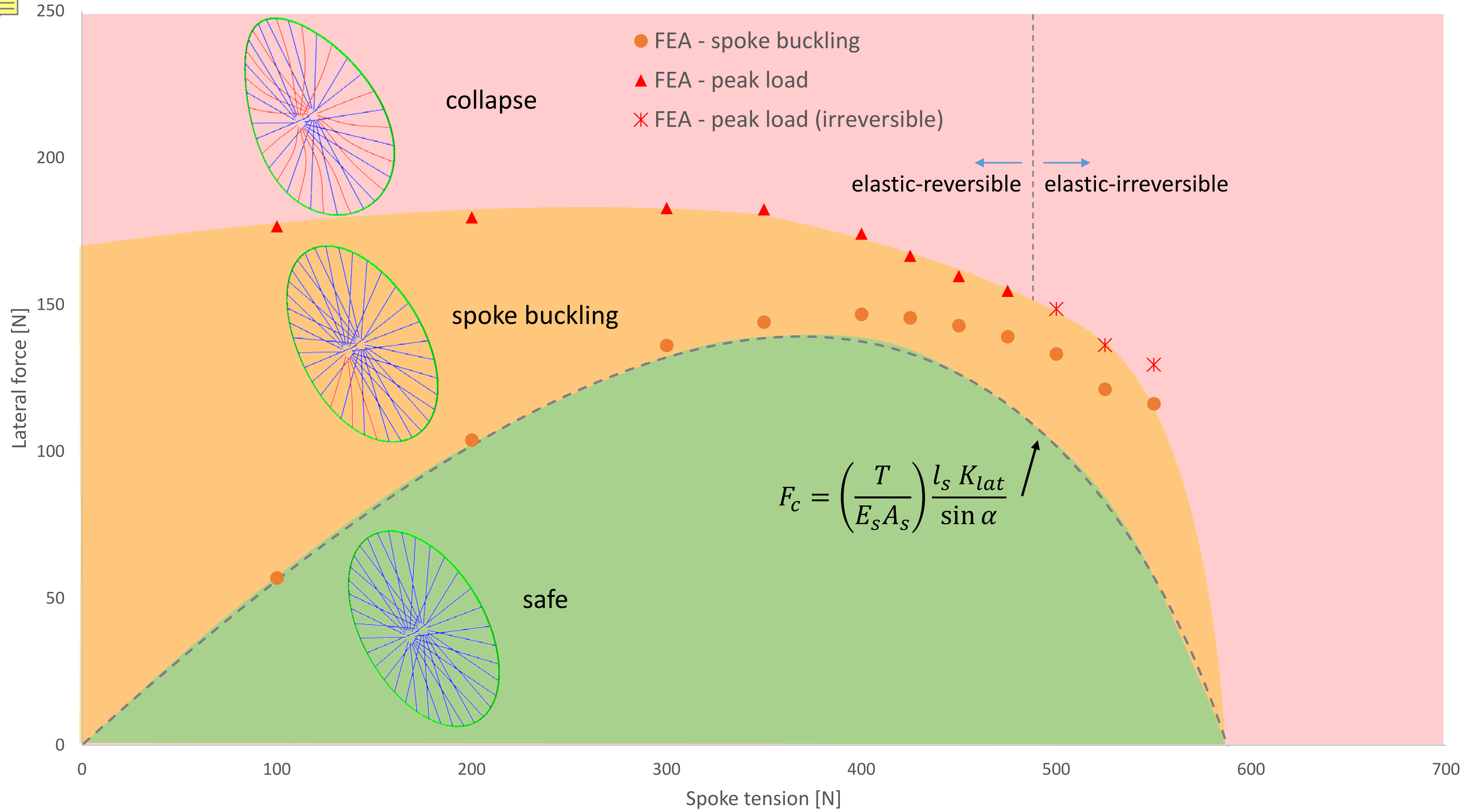
Uniform spoke tension Lateral force Radial force

Buckling under spoke tension



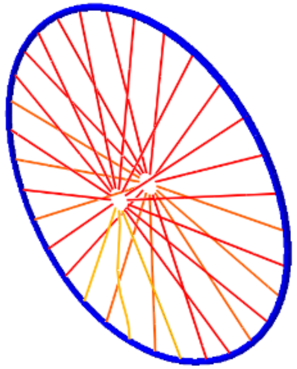


Ford, Papadopoulos, Balogun, Bicyc. Mot. Dyn. Conf. 2016

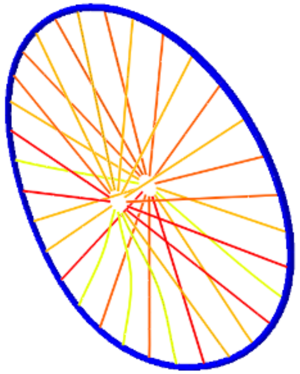


FEA - spoke buckling
FEA - peak load
FEA - peak load (irreversible)
collapse
spoke buckling
safe
elastic-reversible
elastic-irreversible
$F_c = \left(\frac{T}{E_s A_s}\right)\frac{l_s\, K_{lat}}{\sin\alpha}$
Lateral force [N]
Spoke tension [N]
0
50
100
150
200
250
0
100
200
300
400
500
600
700

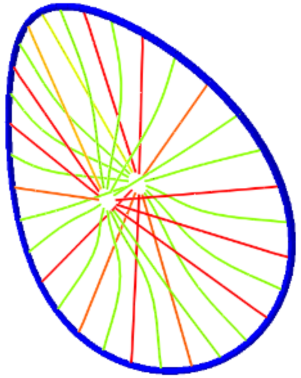
Collapse under radial load



Lower spokes buckle and leave rim unsupported



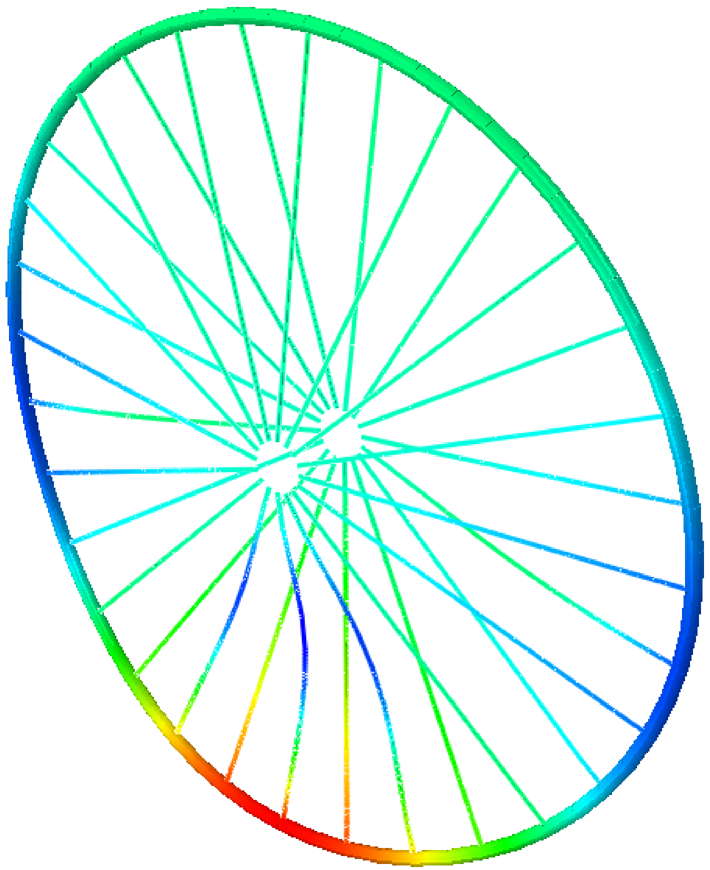
Rim deflects laterally at the load point

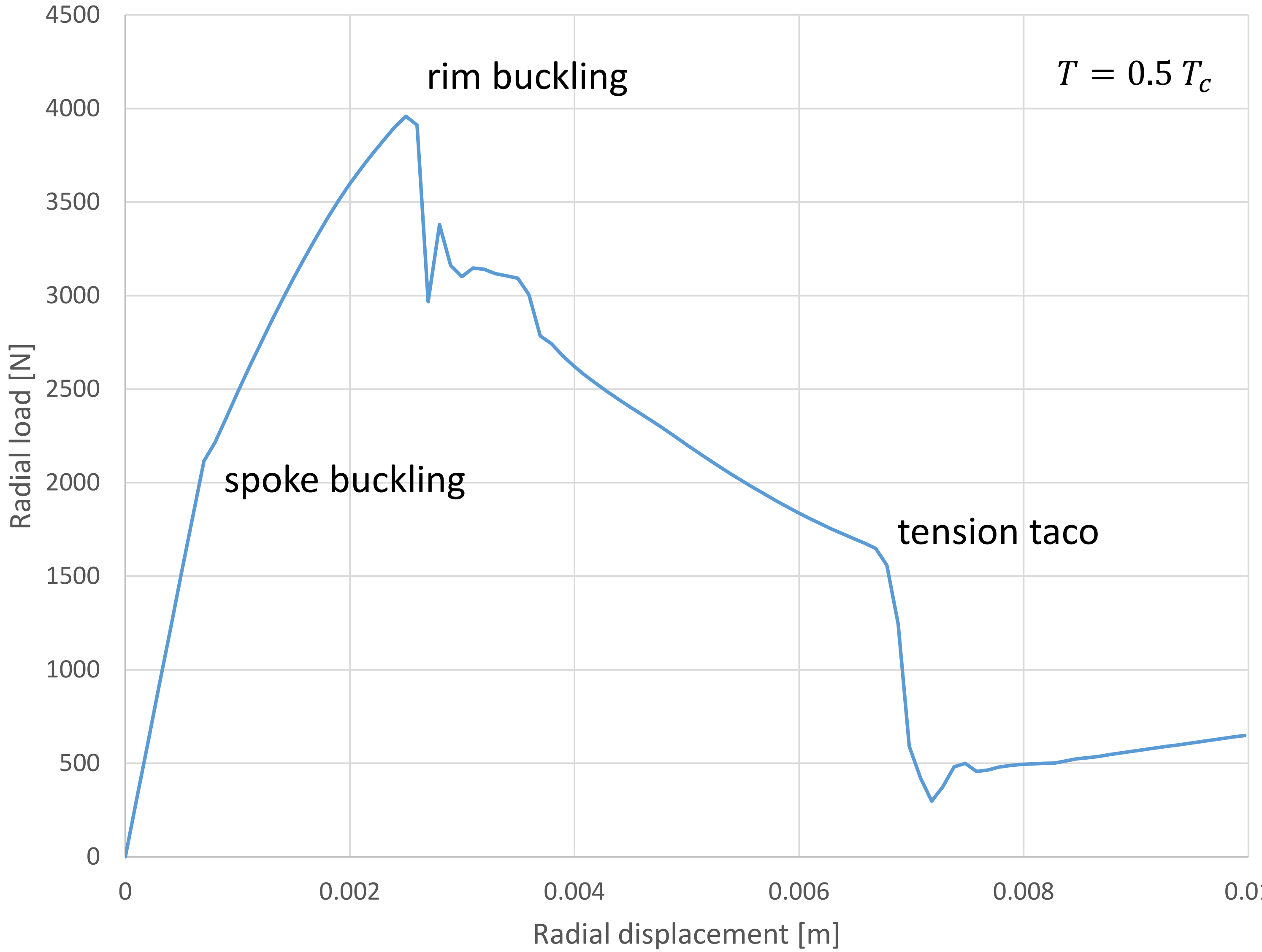


The loss of stiffness causes the rim to collapse under its own tension

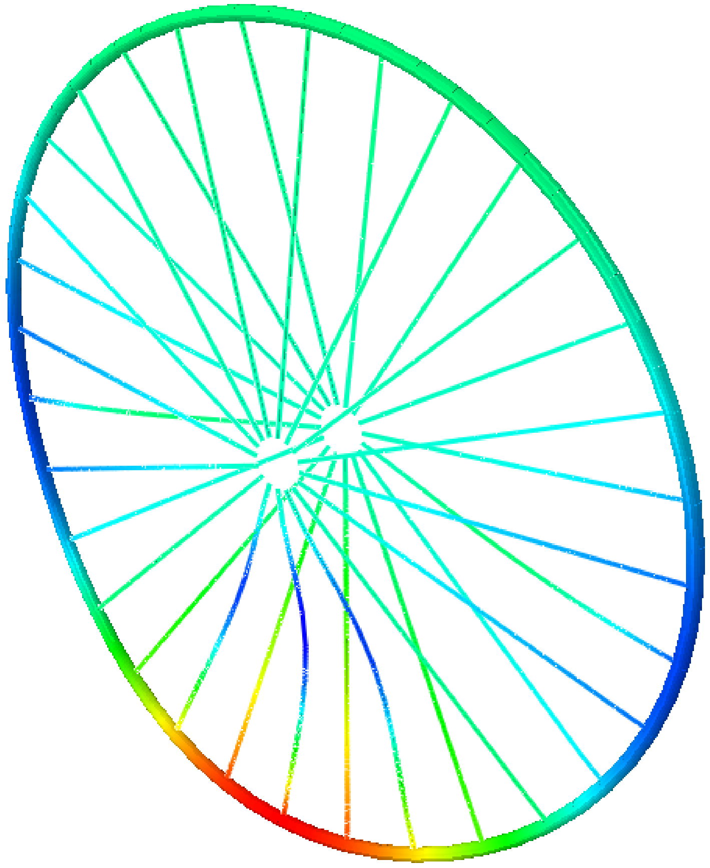
Sean Williams, “Taco Wheel” (youtube.com)

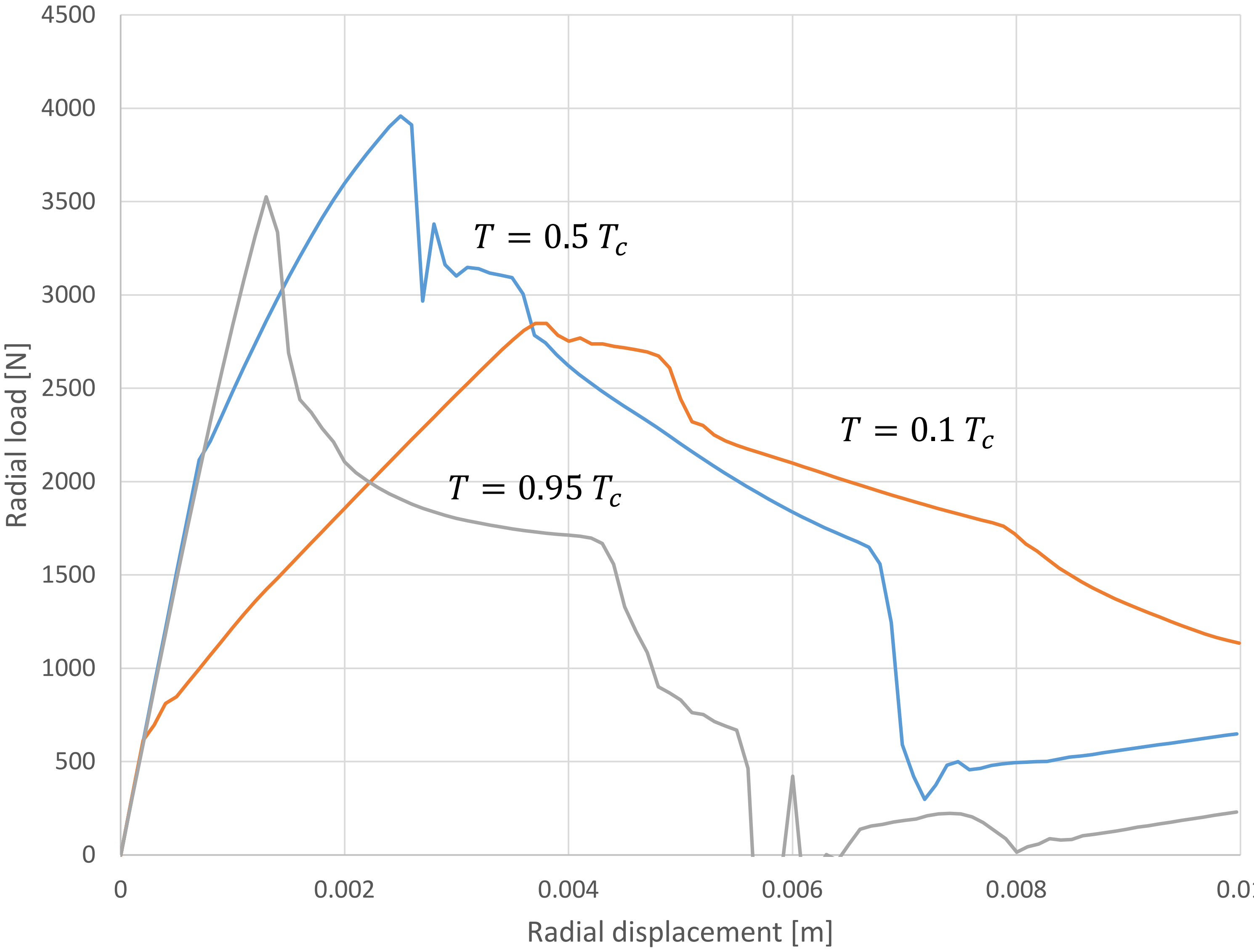
Dual(ing) role of spoke tension



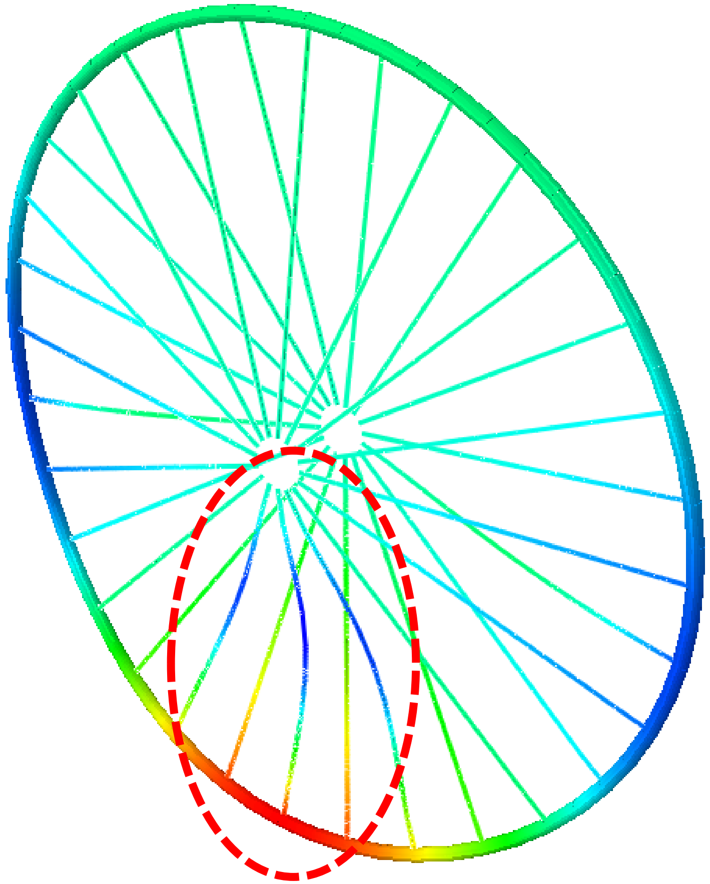


Dual(ing) role of spoke tension



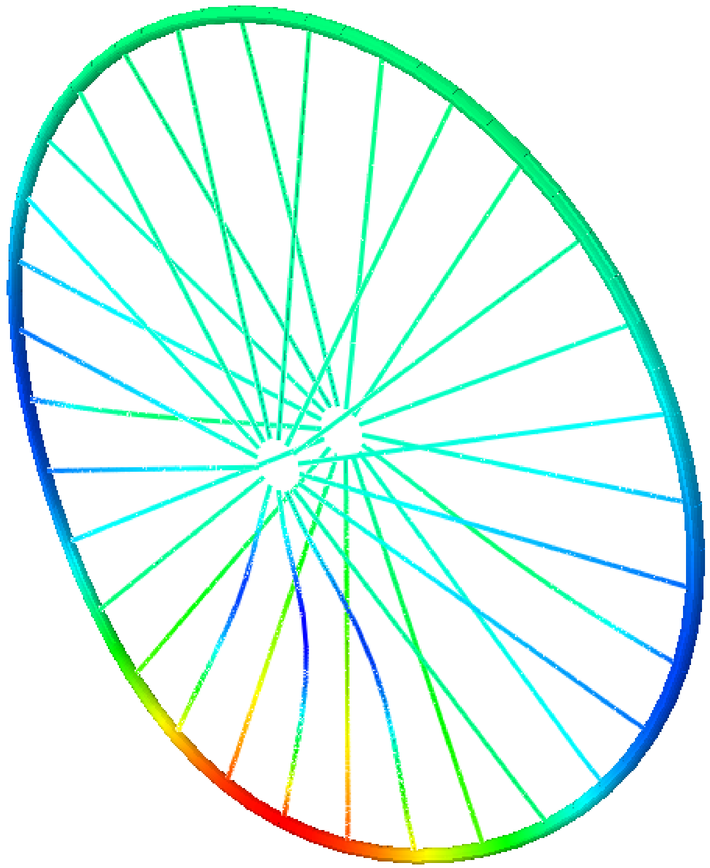


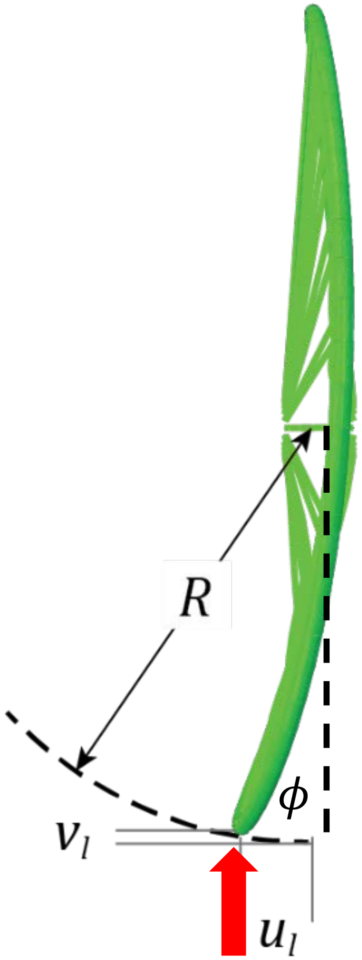
Buckling of the **spokes**

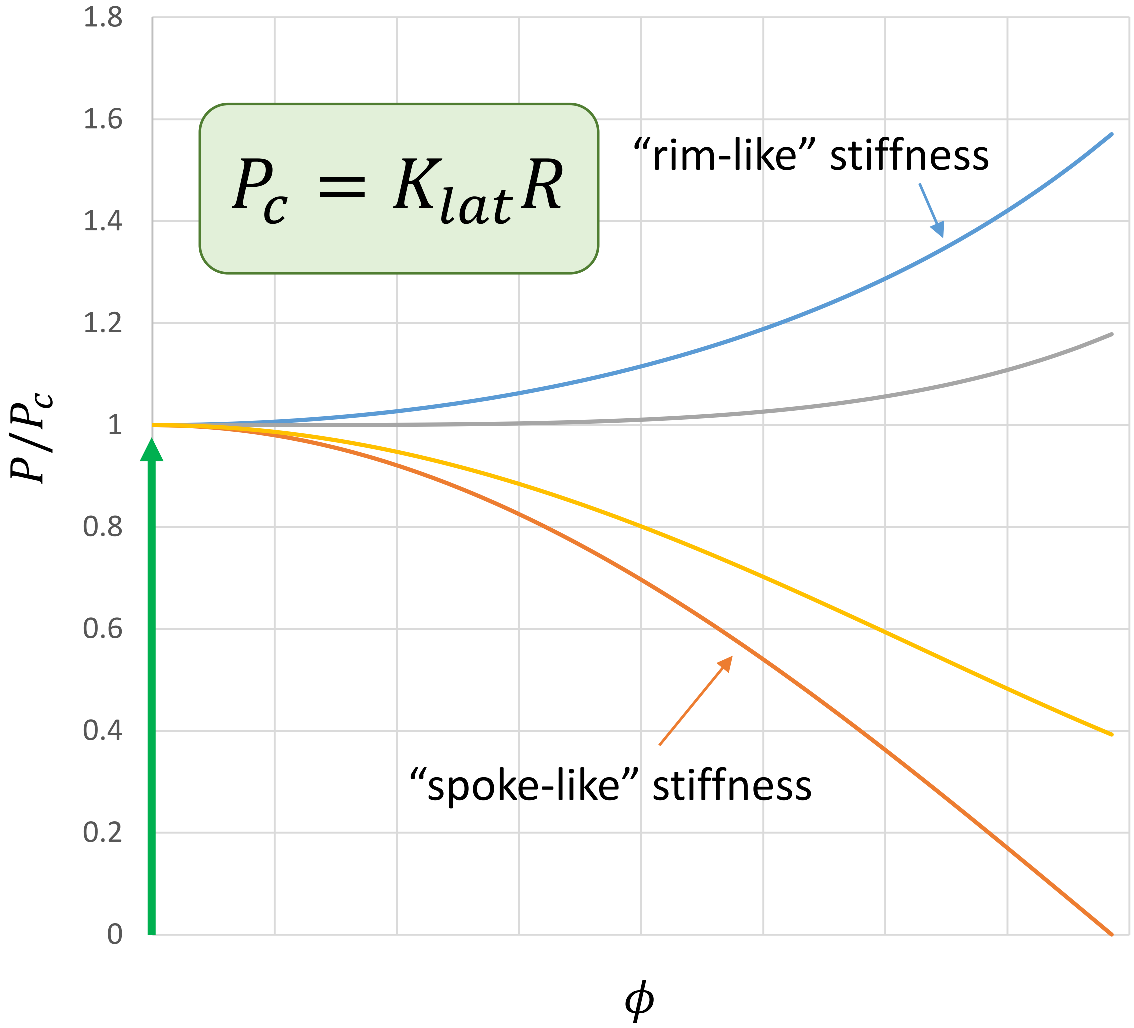


$$P_c = T\left(\frac{K_{rad}}{\cos\alpha}\right)\left(\frac{l_s}{EA}\right)$$

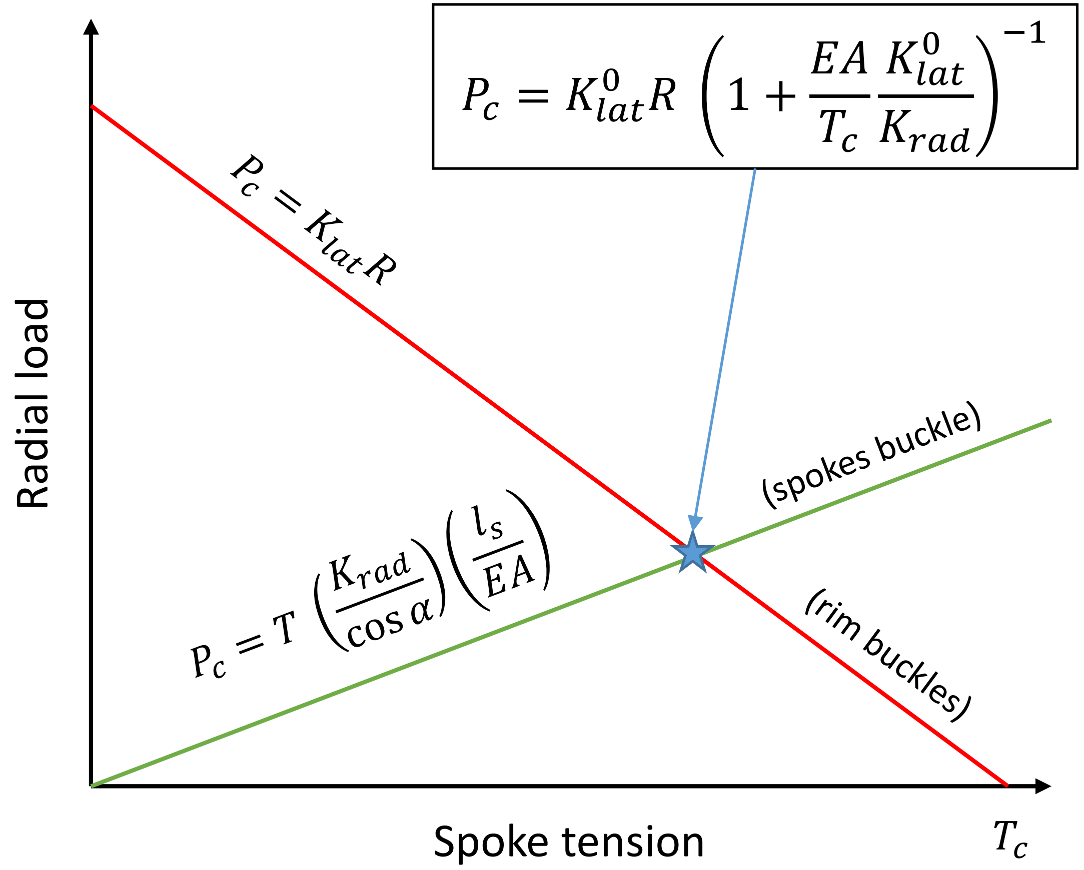
Buckling of the **rim**

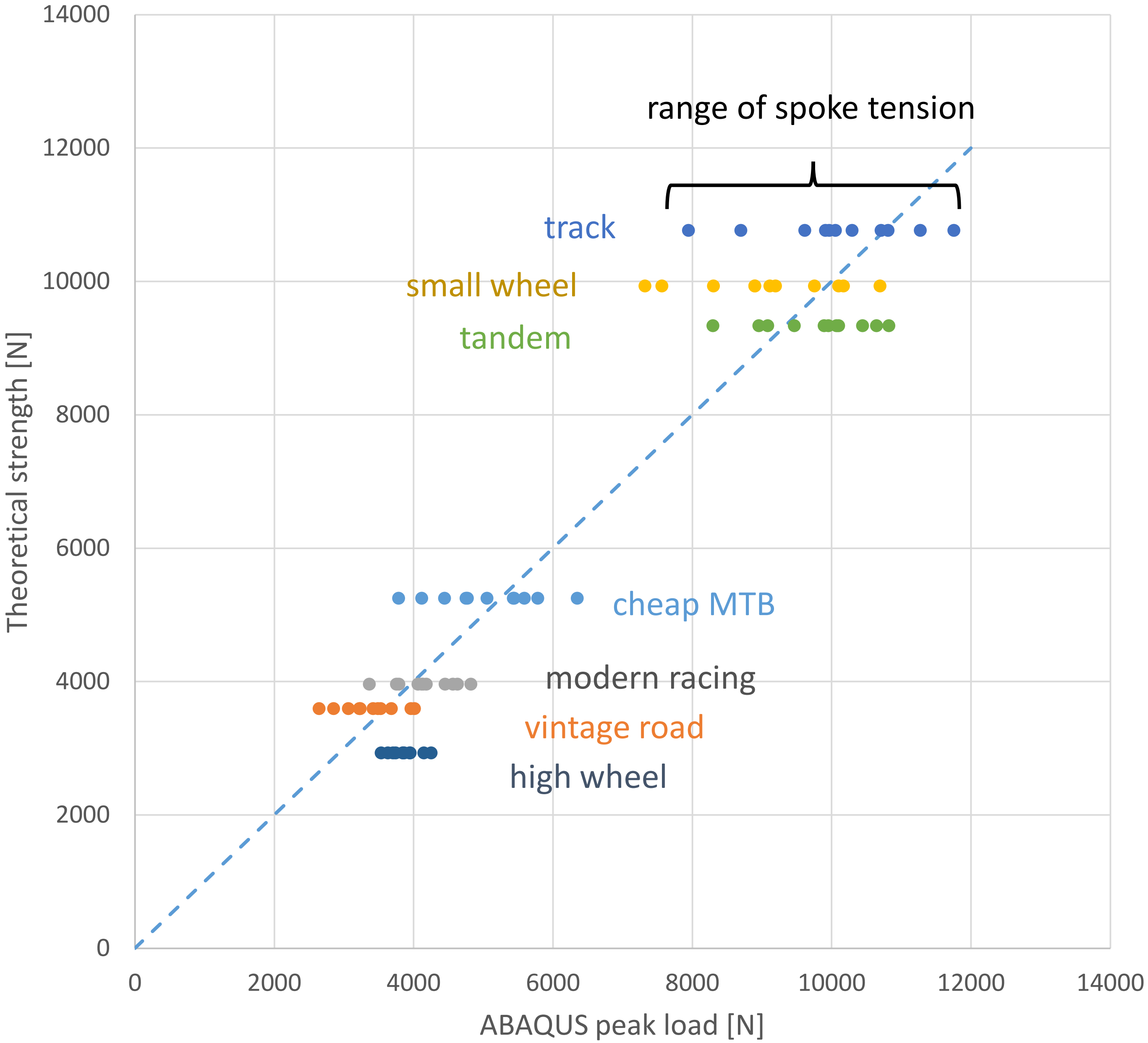




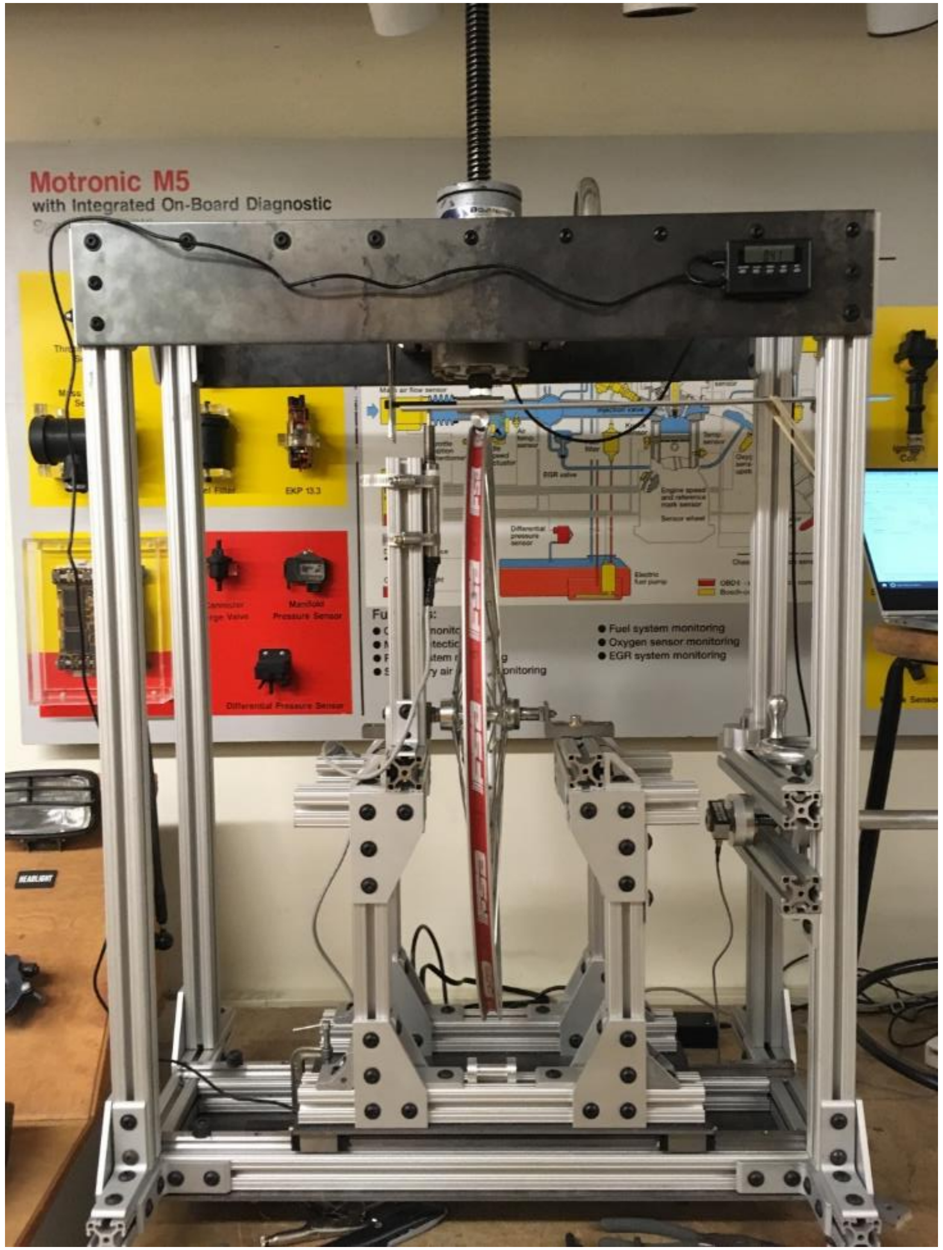


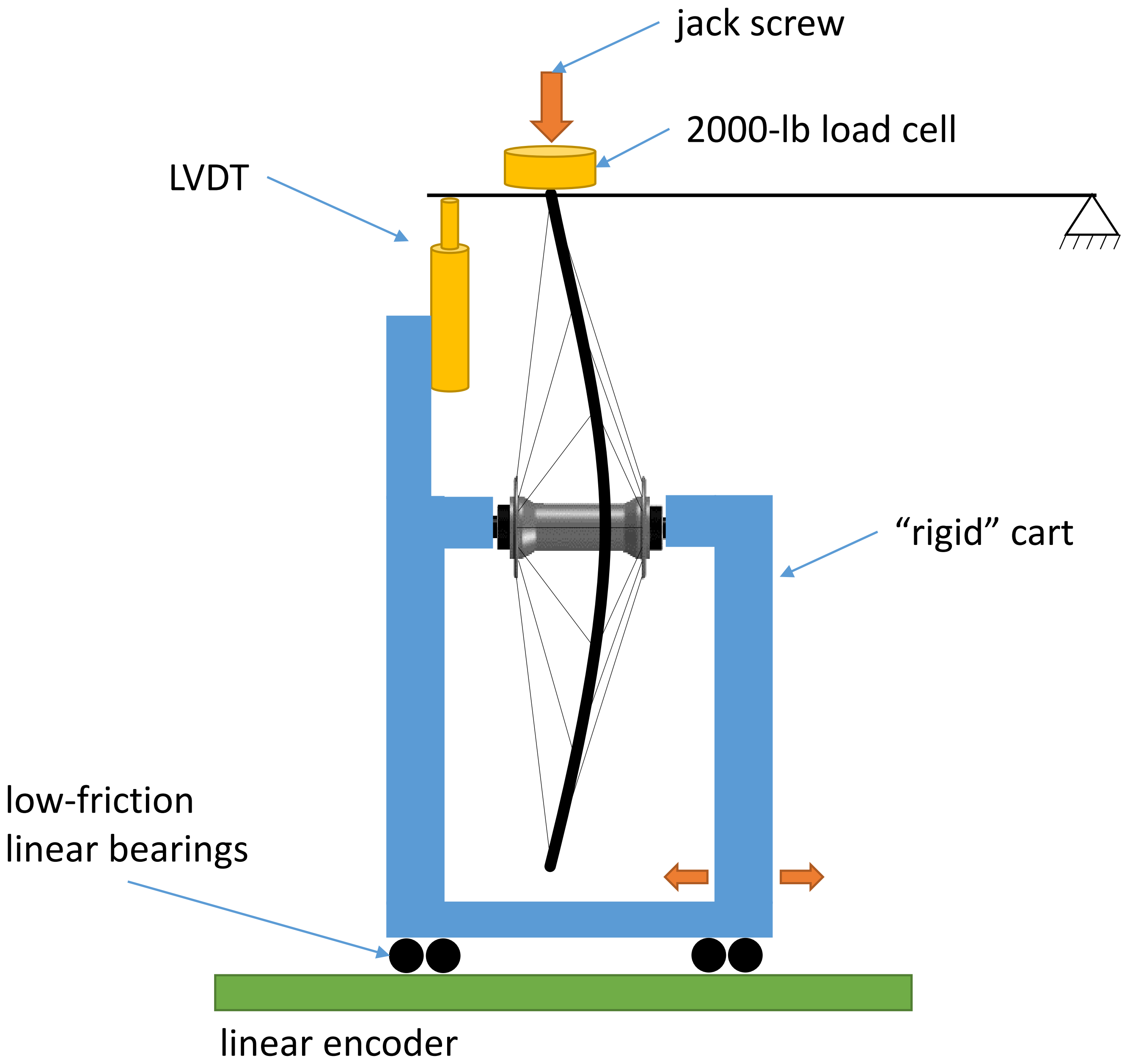
Rim + Spokes buckling





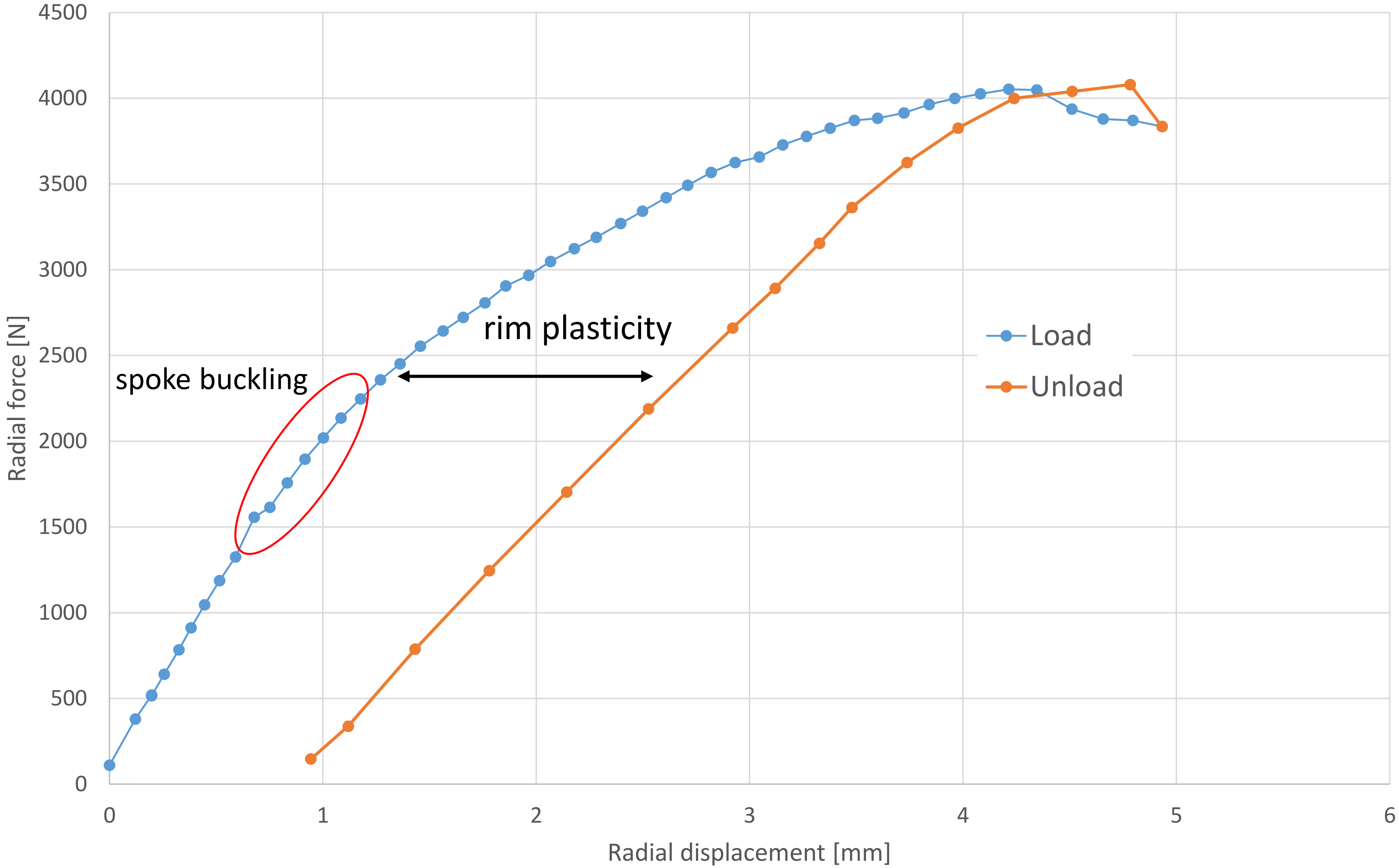
NEU Wheel Crusher



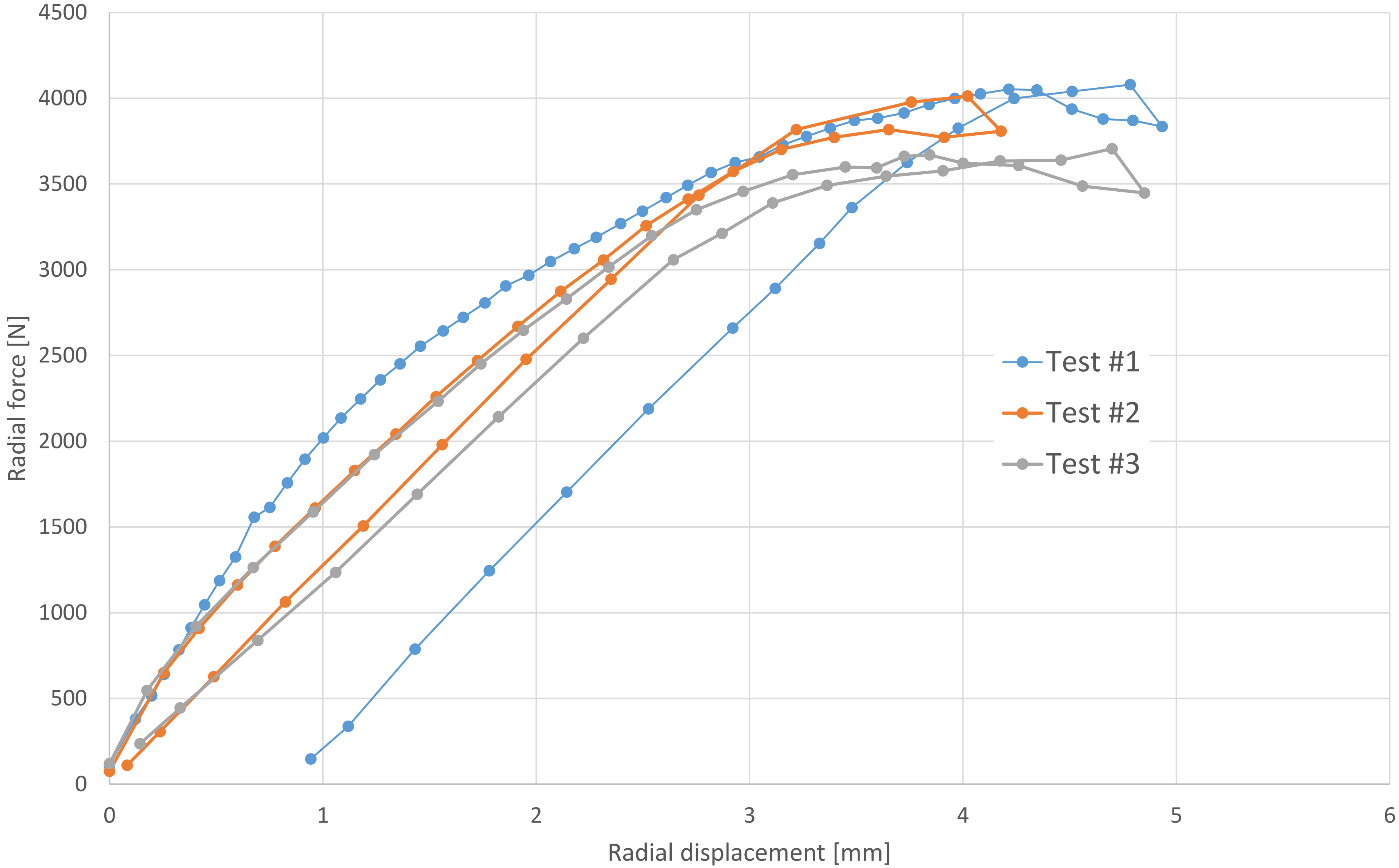


Developed by Northeastern University Capstone Team (see acknowledgements slide)

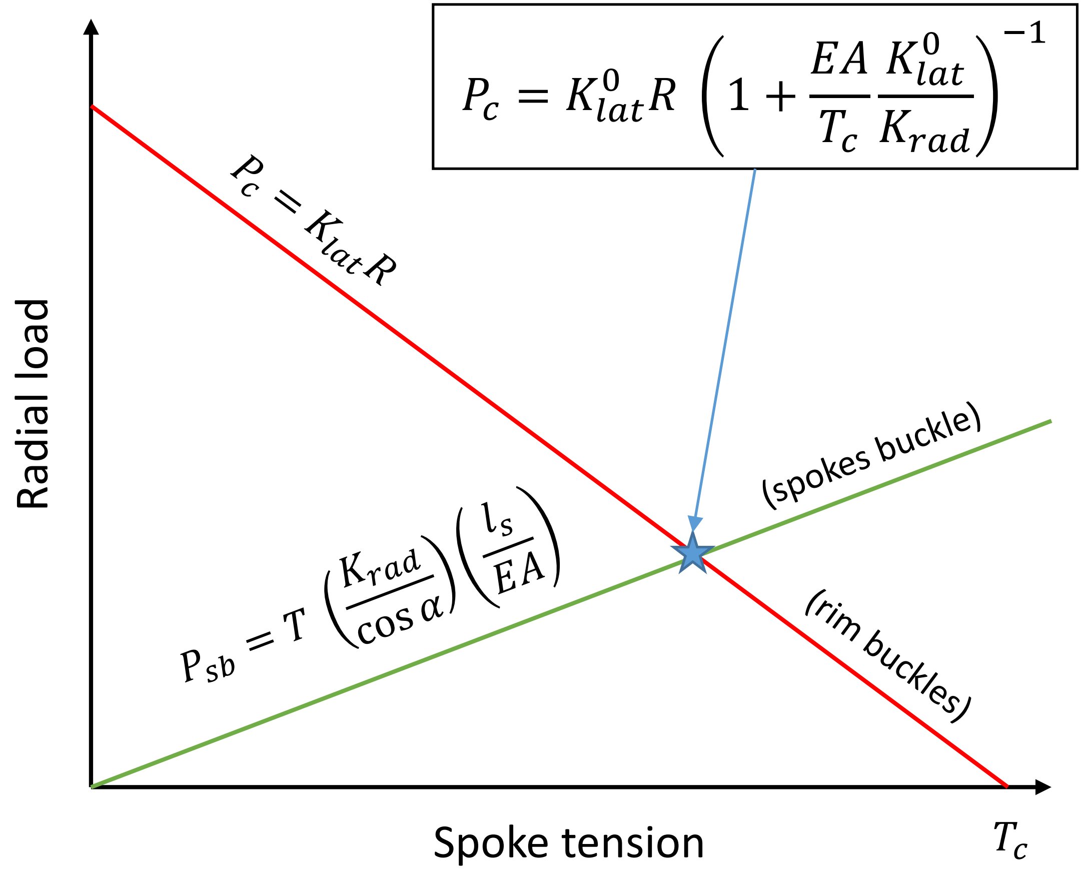
Buckling under radial load

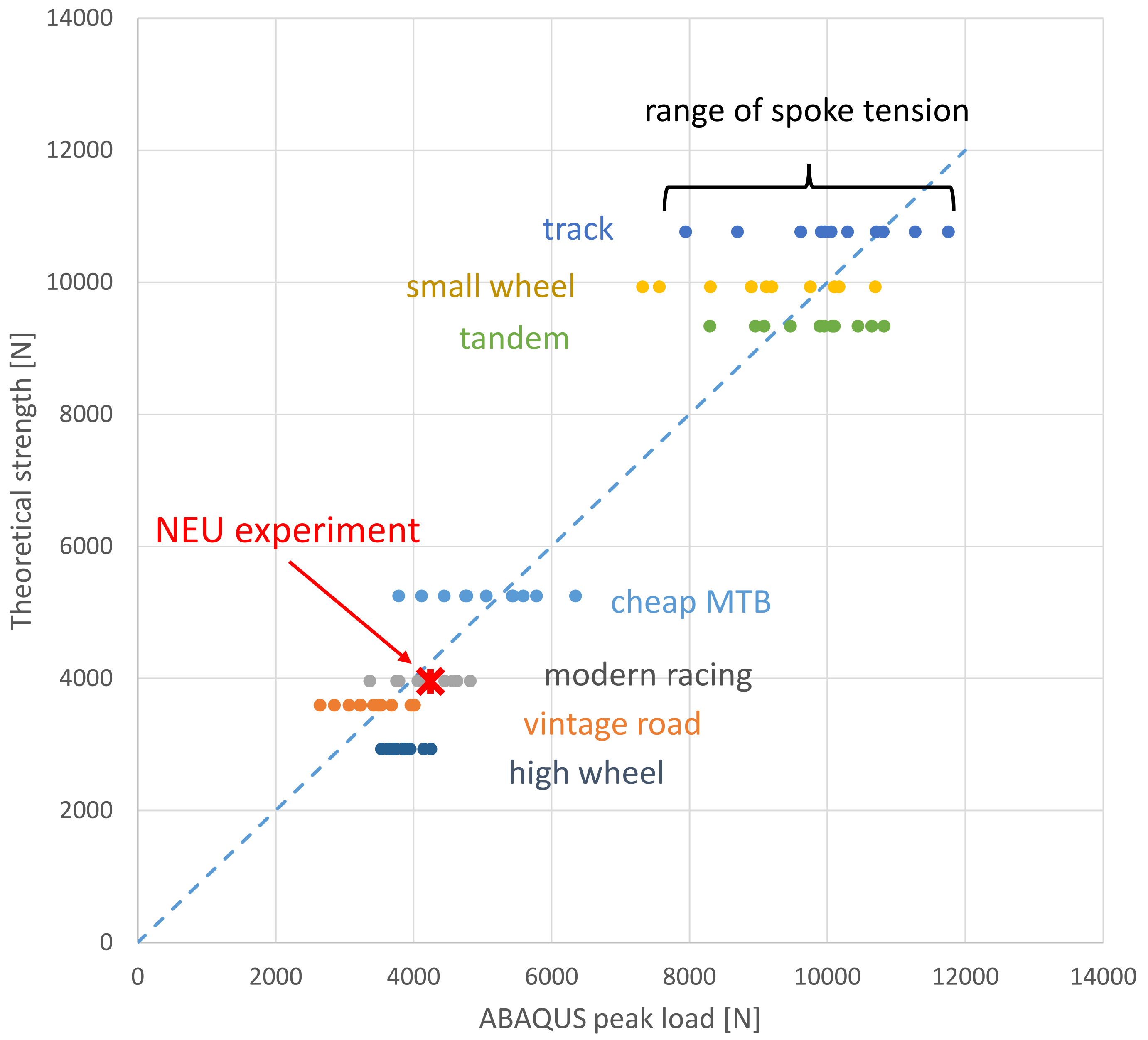


Buckling under radial load



Model comparison



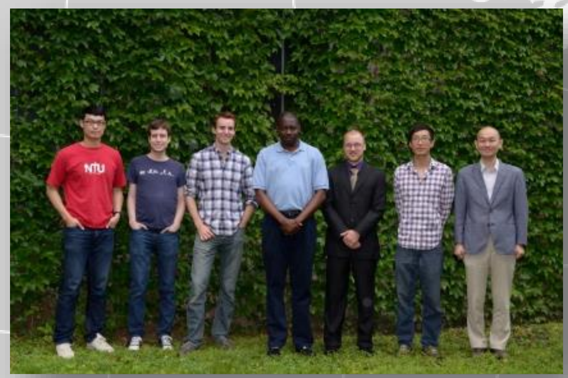


BOEING

Patrick Peng

Northwestern
McCORMICK SCHOOL
OF ENGINEERING

the recyclery



Oluwaseyi Balogun

John Rudnicki, Isaac Daniel

NU Transportation Library

The Recyclery Collective

Northeastern University
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